



AVVA SAI PRANAV

3RD YEAR B.TECH
MECHANICAL

CONTACT

MOBILE NUMBER: 9611635599

EMAIL ID: avva.saipranav@gmail.com

GITHUB: <https://github.com/avvasai>

EDUCATION

P.E.S UNIVERSITY

AUGUST 2018-PRESENT

- BACHELOR OF TECHNOLOGY IN MECHANICAL ENGINEERING
- CURRENT CGPA: 7.49

NATIONAL PUBLIC SCHOOL

MAY 2017-MAY 2018

- CBSE: CLASS 12TH
- OVERALL PERCENTAGE: 87%

NATIONAL PUBLIC SCHOOL

MAY 2015- MAY 2016

- CBSE: 10TH
- OVERALL GPA : 9.6

PROFILE

An engineer aspirant with a keen sense of problem-solving and solution implementation. A wide array of skill set nurtured by a never-ending enthusiasm and eagerness to learn. Wanting to make a difference with my research work for the needy. Given the opportunity would like to work on affordable, technologically advanced artificial limbs and mechanical systems.

EXPERIENCE

TEAM HAYA RACING:

STUDENT RACING CHAPTER OF P.E.S

VEHICLE DYNAMICS DOMAIN

- Design of equipment for Design Validation: Verification of design using physical sensors and measurements.
- Tire Data Analysis: Analysis of different tire compounds and size, determining the most suitable tire.
- Suspension: It ranges from analysis of hardpoints to the modelling of A-arms and rockers.

STUDENT MENTOR FOR PROJECT IGNITION

- Mentoring 1st year students in an internship on the basics of vehicle dynamics and data acquisition systems.

RESEARCH PROJECTS

BIOMIMETICS OF HUMMINGBIRD: ANALYSIS OF WING AND KINEMATIC DESIGN

- The motion of the humming bird was analyzed using image processing techniques.
- Weight estimation was conducted after which wing design is simulated. Once a wing was found that could lift the estimated weight. We calculated the lift for various flapping frequencies
- Kinematic design is generated using the wing specifications and the motion of the humming birds which was analyzed.

INTERNSHIP

INTERNSHIP UNDER A PROFESSOR PROGRAM

- Currently working under Professor Sethuram on the topic of mechatronics and robotics.
- Tasked designing machines for the sorting of blood samples.
- Working on two approaches. One being a conveyor belt system and the second being a 4DOF robotic arm.

TEAM HAYA INTERNSHIP: WIRELESS STEERING WHEEL DISPLAY

- It was held in the summer of 1st year as a program under the racing team for members not a part of the team.
- Designed a wireless steering wheel
- MQTT protocol facilitated the communication between Raspberry Pi and NodeMCU.
- A Java-based GUI was developed on Processing and was used for the display.

SKILLS

- SOLID WORKS
- ANSYS
- MATLAB
- PROFICIENT IN USE OF POWER TOOLS
- MICRO CONTROLLER (ARDUINO, NODEMCU & EYFI MEGA)
- LINUX
- PYTHON
- WEB DEVELOPMENT
- PROCESSING(JAVA BASED IDE)
- MACHINE LEARNING
- IMAGE PROCESSING TECHNIQUES

LANGUAGES

- ENGLISH- READING & WRITING
- HINDI - READING & WRITING
- TELUGU - BASIC COMMUNICATION
- KANNADA- BASSIC COMMUNICATION
- JAPANESE- PREPARING FOR N5 LEVEL

SOFT SKILLS

- EXCELLENT PRESENTATION SKILLS
- THE ABILITY TO ANALYZE COMPLEX TECHNICAL INFORMATION
- THE ABILITY TO RESEARCH EFFECIENTLY
- EXPERIENCED IN THE PROCESS OF SELF-LEARNING

PROJECTS

1. Computational Design of Machine Elements

- Simplified the manual calculations of the design of machine elements using Matlab.
- The design procedure and values were cross-verified using a Data Handbook.
- The implemented machine elements are shafts, springs, springs, bevel gear, helical and worm gears.

2. Design of Data Acquisition system

- Designing a compact data acquisition for measuring vehicle dynamic parameters
- The data gathered from such a system would be used for design validation and vehicle performance tuning.

3. Rapid Rescuer

- Implemented a solution for rescuing people and relief materials in case of disaster
- Used OpenCV for identification and SLAM algorithm for path planning.
- Used python-based socket communication for remote robot operation.

NOTABLE ACHIVEMENTS

- Thirty-six hours of social service in the National Library of Singapore.
- Recipient of "The most helpful student award" on two occasions.
- Commendable project award in a 1st-year internship.
- High commendation award in Model United Nations in YLGC Coimbatore
- Placed 1st in the Most Extreme Challenge of PES under the event of Aatmatrishah.
- Ranked 2nd in Rube Goldberg competition in the annual science fest of PES University called Equinox.

POSITION OF RESPONSIBILITY

AMATEUR SCIENTIST

- A college hosted event for school students. Consisted of a variety of competitions
- Volunteer for the Art Department in 1st Year.

AATMATRISHA

- The flagship college fest of PES University.
- Volunteer for the Art Department in 1st and 2nd Year.

DIPLOMAT WARS

- A chapter under the MUN society of PES University. The competition had multiple rounds where students were tasked to go through debates and quizzes on the current world affairs.
- Core Organizer for the Quiz Round.

MODEL UNITED NATIONS

- Mentoring high school students on how MUN's work and the procedure for participating in a MUN.
- Written articles for the official Instagram handle and website under the initiative known as "For the Record"